AI-Driven Anomaly Detection

for Cyber Threats in Smart Grid Infrastructure

Using TON_IoT Network Logs

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Why Smart Grid Cybersecurity Matters



Expanding Attack Surface

IIoT connectivity links critical infrastructure to global networks — every new device is a new entry point.



100% Availability Required

Smart grids cannot tolerate downtime. DDoS attacks directly threaten power delivery to millions.



IDS Blind Spots

Traditional signature-based systems cannot detect zero-day or evolving attack patterns.

Key Challenge: Detecting unknown attacks in high-volume, real-time network traffic

Current Limitations & Research Gap

Existing Approach	Core Limitation
Signature-Based IDS	Cannot detect zero-day or previously unseen attacks
Supervised ML	Requires large amounts of labeled attack data
Rule-Based Systems	Static — cannot adapt to evolving threat landscapes

Research Gap

Need for unsupervised anomaly detection methods capable of identifying unknown attacks in IIoT environments — without relying on labeled attack data.

Research Questions



RQ1

Can unsupervised anomaly detection effectively detect DDoS attacks in smart grid network environments?



RQ2

How does Extreme Learning Machine compare with Isolation Forest and Autoencoder approaches?



RQ3

What trade-offs exist between detection accuracy and false positive rates?

Dataset: TON_IoT & Research Pipeline

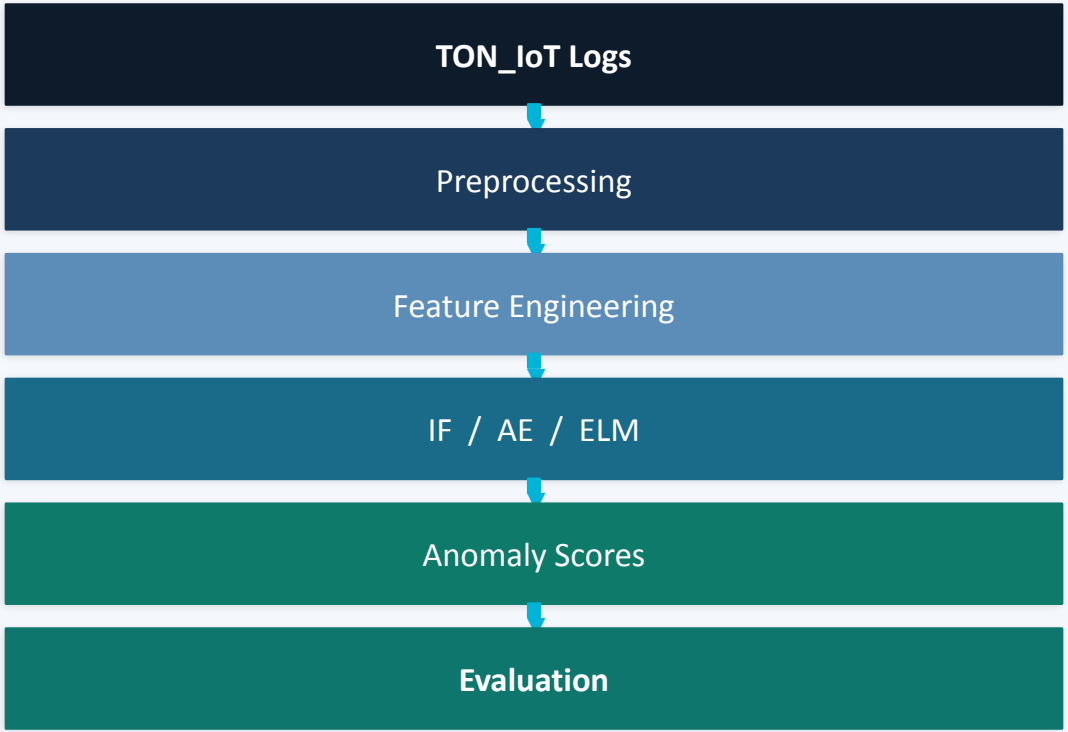
TON_IoT Dataset

~37,000
Network Records

Zeek/Bro
conn.log format

40–45
Features

80/20
Train / Test Split



Dataset & Pipeline — Labels used ONLY for evaluation, not training

Comparative Methodology



Isolation Forest

Unsupervised (Baseline)

Fast, scalable anomaly detection via feature-space isolation. No labels needed.



Autoencoder

Unsupervised Deep Learning

Learns compressed normal-traffic representations. High reconstruction error flags anomalies.

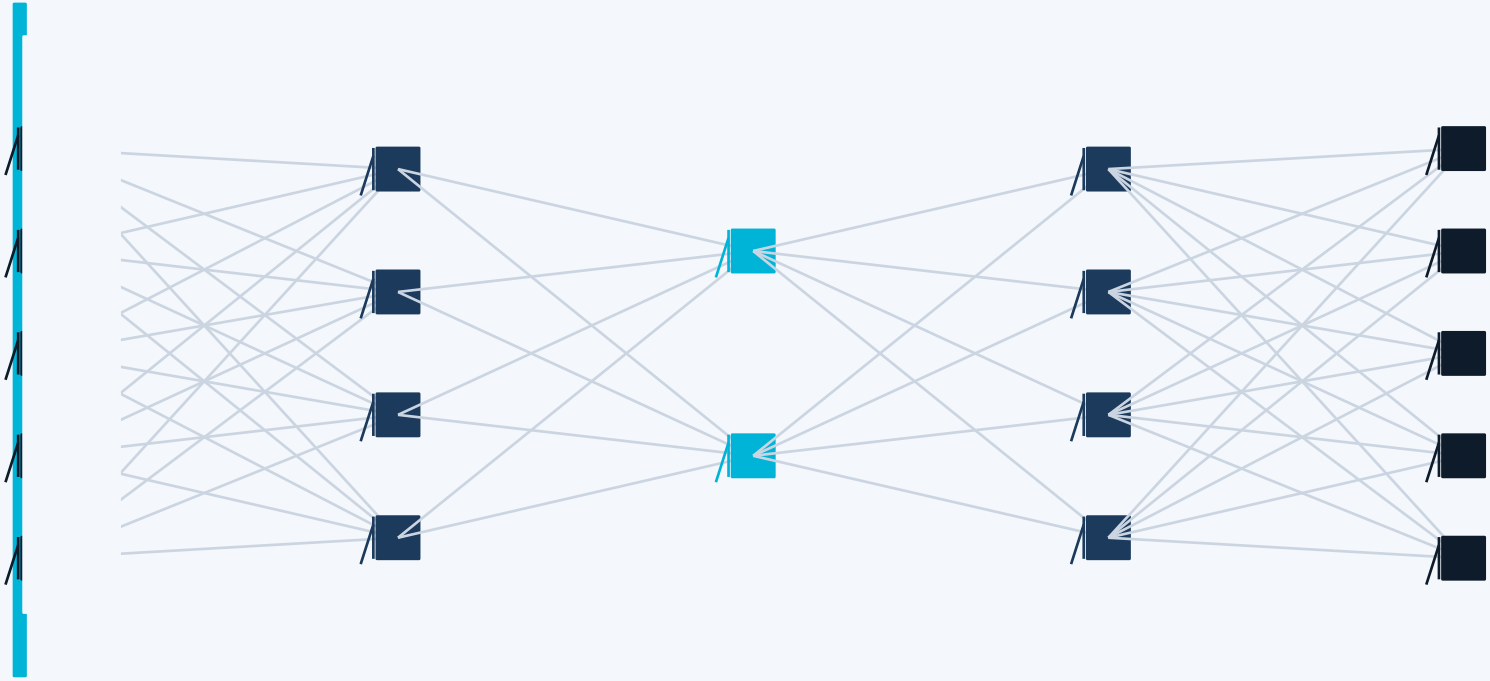


Extreme Learning Machine (ELM)

Supervised

Analytical weight computation — extremely fast. Requires labeled data for training.

Autoencoder Mechanism



Input Features

Encoder 64 → 32

← ENCODER →

Latent Space×16

Decoder 32 → 64

← DECODER →

Output (Reconstructed)

Key Principle: High reconstruction error = deviation from learned normal patterns = potential anomaly

Autoencoder — Trained exclusively on normal traffic

Experimental Setup & Protocol

Data Split

80 / 20 train-test split — StandardScaler normalization

Unsupervised Training

Autoencoder trained on normal traffic only — no labels used during learning

Detection Strategy

Percentile-based thresholds (95%, 90%, 85%, 80%) applied to reconstruction error

Evaluation Metrics

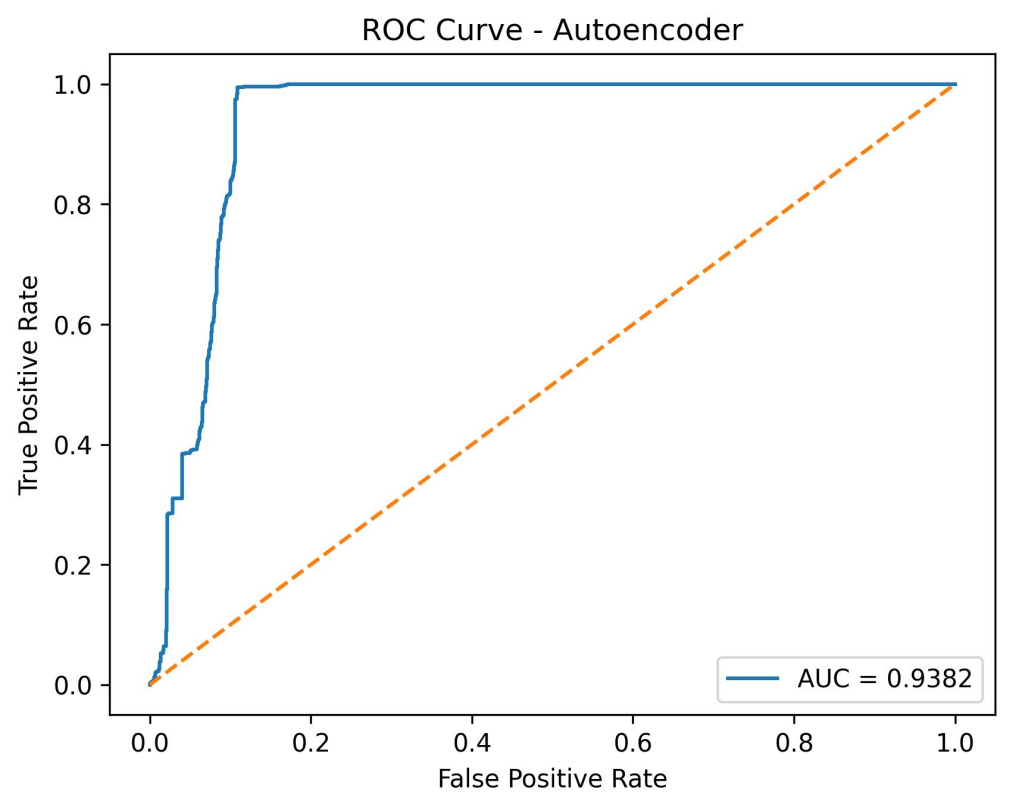
Precision · Recall · F1-score · ROC-AUC — evaluated against ground truth labels

Autoencoder Performance Results

ROC-AUC

≈ 0.94

- Strong separation between normal & anomalous traffic
- High precision at conservative thresholds (Precision ≈ 0.97)
- Recall governed by threshold selection — not model capability



Results — Model separates normal vs anomalous traffic across all thresholds

Operational Threshold Sensitivity

Threshold	Precision	Recall	Use Case
95% (Conservative)	0.97 ✔ Very High	0.39 ✗ Low	Minimize false alarms
90% (Moderate)	0.97 ✔ High	0.83 ✔ Good	Balanced deployment
85% (Aggressive)	0.97 ✔ High	0.99 ✔ Very High	Maximum threat detection
80% (Sensitive)	0.95 ✔ High	0.99 ✔ Very High	Security-critical systems

Key Insight

Threshold selection governs operational performance — not model architecture. The same model can be tuned from low-false-alarm precision mode to maximum-recall detection mode.

Comparative Analysis

Model	Precision	Recall	F1
Isolation Forest	0.94	0.91	0.92
Autoencoder	0.97	0.83	0.90
ELM	0.99	0.99	0.99

Model	Strength	Limitation
Isolation Forest	Fast & scalable	Simpler patterns
Autoencoder	Complex nonlinear learning	Threshold sensitive
ELM	Highest accuracy	Needs labeled data

Model Agreement

Isolation Forest & Autoencoder agree on 94% of cases — confirming consistent anomaly identification. Their disagreements reveal complementary sensitivity.

Conclusions on Research Questions

RQ1 ✓

Unsupervised models effectively identified DDoS anomalies

ROC-AUC ≈ 0.94 confirms strong discriminative capability without labeled training data.

RQ2 ✓

ELM outperformed in supervised settings

Precision/Recall/F1 ≈ 0.99, but performance depends entirely on labeled data availability.

RQ3 ✓

Operational trade-offs are controllable via threshold tuning

Conservative thresholds → high precision. Aggressive thresholds → high recall. Both achievable.

Research Questions — All three answered with empirical evidence

Research Limitations

Dataset Scope

Evaluation on single dataset (TON_IoT) — generalizability to other IIoT environments remains to be tested.

Validation

No cross-validation or multi-run statistical analysis — reproducibility relies on fixed random seeds.

Deployment

No real-time operational testing — static dataset analysis only.

Threshold Tuning

High sensitivity to manual threshold selection — adaptive strategies not yet implemented.

Future Research Directions



Advanced AI

LSTM & Graph Neural Networks for temporal and relational attack patterns



Explainability

XAI techniques to explain anomaly causes for SOC analysts



Automa

Adaptive auto-t
for dynamic r
environm



Real-World Deployment

Live SOC integration and multi-dataset generalization testing



Federated Learning

Privacy-preserving distributed anomaly detection across IIoT networks

The foundation is set.
Next step: deploy, adapt, and explain.

Future systems will combine adaptive thresholding, temporal deep learning, and explainable AI for robust real-world IIoT security.

Conclusion

Machine learning-based anomaly detection demonstrates strong potential for identifying cyber threats in smart grid environments where labeled data is limited.

- Unsupervised detection works — ROC-AUC ≈ 0.94
- Threshold tuning is the key operational lever
- Hybrid approaches offer the strongest coverage

Thank you — Questions?